INCH-POUND
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# DETAIL SPECIFICATION

# PRINTED WIRING ASSEMBLY, I/O

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers an Input/Output (I/O) Printed Wiring Assembly, herein referred to as the PWA. This is one of six printed wiring assemblies which comprise the Simplified Test Equipment/Internal Combustion Engine-Reprogrammable (STE/ICE-R) Vehicle Test Meter (VTM) (see Drawing 12259265).

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5998

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines (see 4.2.1).

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

12259253 - Printed Wiring Assembly, I/O.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/IPC J-STD-001 - Requirements for Soldered Electrical and Electronic Assemblies.



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(Application for copies should be addressed to the American National Standards Institute (ANSI, 11 West 42nd Street, New York, NY 10036.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA) RECOMMENDED STANDARDS

| | |
|---|---|
| 232 | - Interface Between Data Terminal Equipment and Data Circuit Terminating Equipment Employing Serial Binary Data Interchange. |
| 485 | - Electrical Characteristics of Generators and Receivers for Use in Balanced Digital Multipoint Systems. |

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street NW, Washington, DC 20006.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample PWA shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Design, materials, and manufacturing processes. Design, materials, and manufacturing process selection shall be as specified herein and in applicable referenced specifications, standards, and drawings. Materials shall be uniform and free from imperfections or defects which affect their performance and serviceability. All metallic parts shall be made from corrosion resistant metals or treated with corrosion-resistant materials. Asbestos and Cadmium materials shall not be used in any form in any part of the PWA. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcuries per gram or activity per item equals or exceeds 0.01 microcuries.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Operating requirements. Each PWA shall provide the following functional, operational, and performance capabilities.



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3.3.1 Input power. The PWA shall require not more than the current specified in table I.

TABLE I. Input power.

| Source | Voltage | Current |
|---|---|---|
| P1-5, 6 | 5 Vdc 1/ | 150 mA 2/ |
| P1-13, 14 | 15 Vdc | 7.5 mA |
| P1-19, 20 | -15 Vdc | 8.5 mA |

1/ Vdc = Volts direct current
2/ mA = milliampere

3.3.2 Logic levels.

3.3.2.1 Complimentary metal oxide semiconductor (CMOS) logic levels. Unless otherwise specified (see 6.2), CMOS logic levels accepted by and output from the PWA shall be as follows:

Input High (Logic Level 1) = 3.5 Vdc minimum (min)
Input Low (Logic Level 0) = 1.5 Vdc maximum (max.)
Output High (Logic Level 1) = 4.5 Vdc min
Output Low (Logic Level 0) = 0.5 Vdc max.

3.3.2.2 Transistor-Transistor Logic (TTL) logic levels. Unless otherwise specified (see 6.2), TTL logic levels accepted by U16 and U17 shall be:

Input High (Logic Level 1) = 2.0 Vdc min
Input Low (Logic Level 0) = 0.8 Vdc max.

3.3.3 Input isolation. (NOTE: A suffix "/" following a capitalized signal name denotes logic negation.)

3.3.3.1 Data bus isolation. Logic levels applied to any one BUS pin (P1-21 through P1-28) shall not be affected by logic levels present at any other BUS pin.

3.3.3.2 J3-T, J3-U, IOF/7, MRD/, OEN-Q, OEN-Q/, Q, CLEAR/, and TPB isolation. Logic levels applied to any one of P1-11/P1-12, P1-15/P1-16, P1-35 through P1-38, P1-57, and P1-66 shall have no effect on, nor be affected by logic levels applied at any of the other pins specified in this paragraph. P1-62 should be grounded through a 1 kilohm (Kohm) resistor to yield appropriate results.

3.3.4 DCA/-15 Vdc source. The voltage at P1-19 and P1-20 shall not be greater than 0.6 Vdc when P1-42 is pulled up to a voltage greater than 0.4 Vdc and less than 32 Vdc.



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3.3.5 Reset.

3.3.5.1 Power-up reset. P1-68, U1 pin 21, U2 pin 2, U3 pin 2, U9 pin 1, U12 pin 13, and U20 pins 1 and 6 shall conform to the timing diagram shown in figure 1 when the voltage at P1-5 and P1-6 is brought from 0 Vdc to +5 Vdc.

3.3.5.2 CLEAR/. Applying a logic level 0 to P1-62 shall bring P1-68 to logic level 0 in less than 62 nanoseconds (ns).

3.3.5.3 States during power-up reset or CLEAR/. While P1-62 is held at logic level 0, the following pins shall be at the logic level designated in table II.

TABLE II. Logic levels during reset or clear.

| Pin | Logic Level |
|---|---|
| P1-73 | 1 |
| P1-76 | 1 |
| P1-79 | 0 |
| U8 pins 3, 5 | 0 |
| U9 pins 2, 5, 7, 10, 12, 15 | 0 |
| U10 pin 10-14 | 0 |
| U10 pin 15 | 1 |
| U11 pin 4 | 1 |
| U12 pins 3, 9, 11 | 0 |
| P1-75 | 1 |
| P1-78 | 1 |
| U1 pins 24, 25 | 1 |

3.3.6 BUS transceiver.

3.3.6.1 Data in. With U4 pin 1 at logic level 0, data shall be applied and the results shall be observed as shown in table III.

3.3.6.2 Extended I/O latch. While U4 pin 1 is at logic level 0, the waveforms shown in figure 2 shall be applied to latch the data shown in table IV in U9.



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TABLE III. Transceiver bus logic levels.

| Applied data P1- | | | | | | | | Observed data U4 pin | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

TABLE IV. Extended I/O latch data.

| Applied data P1- | | | | | | | | Observed data U9 pin | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 15 | 12 | 10 | 07 | 05 | 02 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

3.3.6.3 Extended I/O decoder. While U4 pin 1 is at logic level 0, the waveforms shown in figure 2 shall be applied to latch the data shown in table V into U9 and the results shall be observed at U10.

TABLE V. Extended I/O decoder data.

| Applied data P1- | | | | | | | | Observed data U10 pin | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 10 | 11 | 12 | 13 | 14 |
| X | X | X | X | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| X | X | X | X | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| X | X | X | X | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 |
| X | X | X | X | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 |
| X | X | X | X | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| X | X | X | X | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 |
| X | X | X | X | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| X | X | X | X | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| X | X | X | X | 1 | X | X | X | 0 | 0 | 0 | 0 | 0 |

(X denotes a don't care condition)

3.3.6.4 BUS transceiver direction logic. U4 pin 1 shall conform to the logic shown in table VI.



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TABLE VI. BUS transceiver direction logic.

| Applied data | | | | Observed data |
|---|---|---|---|---|
| P1-35 | P1-36 | U10 pin 14 | U10 pin 13 | U4 pin 1 |
| 0 | X | X | X | 0 |
| X | 0 | X | X | 0 |
| 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 1 |

(X denotes a don't care condition)

3.3.7 Multiply/divide unit (MDU).

3.3.7.1 MDU crystal oscillator. The signal at Y2 pin 8 shall have a frequency of 1 megahertz (MHz) ± 1 kilohertz (kHz). The rise and fall times of the signal at Y2 pin 8 shall be less than 150 ns.

3.3.7.2 MDU chip select 1 (CS1) logic. U1 pin 23 shall conform to the logic shown in table VII.

TABLE VII. MDU select logic levels.

| Applied data | | | Observed data: U1 pin 23, U2 pin 11, and U3 pin 11 |
|---|---|---|---|
| P1-35 | P1-36 | P1-57 | |
| 0 | X | X | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | X | 1 |

3.3.7.3 MDU register selection. The MDU registers shall be selected for either loading or reading by using the timing diagram of figure 2 and the appropriate data from table VIII.

TABLE VIII. Data to select MDU registers.

| Selected MDU Register | Load or Read | Applied data P1- | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 |
| Status | Read | X | X | 1 | 1 | 0 | 0 | 0 | 1 |
| Control | Load | X | X | 1 | 1 | 0 | 0 | 0 | 1 |
| X | Both | X | X | 0 | 0 | 0 | 0 | 0 | 1 |
| Y | Both | X | X | 1 | 0 | 0 | 0 | 0 | 1 |
| Z | Both | X | X | 0 | 1 | 0 | 0 | 0 | 1 |

(X denotes a don't care condition)



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3.3.7.4 MDU register load. After latching the data from table VIII into U9 to select the MDU Control, X, Y, or Z register by applying the waveforms in figure 2, the data from the register selected shall be loaded by applying the waveforms in figure 3.

3.3.7.5 MDU register read. After latching the data from table VIII into U9 to select the MDU Status, X, Y, or Z register by applying the waveforms in figure 2, the data from the register selected shall be read by applying the waveforms in figure 4.

3.3.7.6 Multiply with add. The MDU shall multiply a 16 bit operand that has been loaded into MDU register X by a 16 bit operand that has been loaded into MDU register Z, then add a 16 bit operand that has been loaded into MDU register Y. The MDU shall produce a 32 bit result, the high order 16 bits of which shall be read from the MDU Y register and the low order bits of which shall be read from the MDU Z register, a minimum of 17 microseconds (µs) after the operation is started by loading the MDU control register.

3.3.7.7 Divide with overflow indication. The MDU shall divide a 32 bit operand, the high order 16 bits of which have been loaded into MDU register Y and the low order 16 bits of which have been loaded into MDU register Z, by a 16 bit operand that has been loaded into MDU register X. The MDU shall produce a 16 bit result that shall be read from the MDU Y register and a 16 bit remainder that shall be read from the MDU Z register. This result and any overflow indication in the MDU status register shall be available 17 µs after the operation is started by loading the MDU control register.

3.3.8 Communications.

3.3.8.1 Universal Asynchronous Receiver/Transmitter (UART) crystal oscillator. The signal at Y1 pin 8 shall have a frequency of 2.4576 MHz ± 250 kHz.

3.3.8.2 Crystal frequency division. U5 pins 6, 5, 3, 2, 4, 13, 12, and 14 shall be the frequencies listed in table IX.

TABLE IX. U5 frequencies.

| U5 Pin | Frequency |
|---|---|
| 6 | 307.20 kHz ± 307 Hz 1/ |
| 5 | 153.60 kHz ± 154 Hz |
| 3 | 76.800 kHz ± 77 Hz |
| 2 | 38.400 kHz ± 38 Hz |
| 4 | 19.200 kHz ± 19 Hz |
| 13 | 9.6000 kHz ± 10 Hz |
| 12 | 4.8000 kHz ± 5 Hz |
| 14 | 2.4000 kHz ± 2 Hz |

1/ hertz (Hz).



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3.3.8.3 Frequency select latch. The frequency select latch shall be selected by using the data in table X and applying the waveforms of figure 2. The data in table XI shall then be latched into U7 by applying the waveforms in figure 5 and the results shown in table XI shall be observed at U7 pins 2, 7, and 10.

TABLE X. Data to select the frequency select latch.

| P1- | | | | | | | |
|---|---|---|---|---|---|---|---|
| 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 |
| X | X | X | X | 0 | 1 | 0 | 0 |

(X denotes a don't care condition)

3.3.8.4 Frequency select. After selecting U7 by applying the waveforms in figure 2 with the data in table X, the waveforms shown in figure 5 shall be applied to latch the data shown in table XII into U7. The frequencies shown in the table shall be observed at U6 pin 5.

TABLE XI. Frequency select latch data.

| Applied data P1- | | | | | | | | Observed data U7 pin | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 10 | 7 | 2 |
| X | X | X | X | X | 0 | 0 | 0 | 0 | 0 | 0 |
| X | X | X | X | X | 0 | 0 | 1 | 0 | 0 | 1 |
| X | X | X | X | X | 0 | 1 | 0 | 0 | 1 | 0 |
| X | X | X | X | X | 0 | 1 | 1 | 0 | 1 | 1 |
| X | X | X | X | X | 1 | 0 | 0 | 1 | 0 | 0 |
| X | X | X | X | X | 1 | 0 | 1 | 1 | 0 | 1 |
| X | X | X | X | X | 1 | 1 | 0 | 1 | 1 | 0 |
| X | X | X | X | X | 1 | 1 | 1 | 1 | 1 | 1 |

(X denotes a don't care condition)

TABLE XII. Frequency select data.

| Applied data P1- | | | | | | | | Observed data |
|---|---|---|---|---|---|---|---|---|
| 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | Frequency at U6 pin 5 |
| X | X | X | X | X | 0 | 0 | 0 | 307.20 kHz ± 307 Hz |
| X | X | X | X | X | 0 | 0 | 1 | 153.60 kHz ± 154 Hz |
| X | X | X | X | X | 0 | 1 | 0 | 76.800 kHz ± 77 Hz |
| X | X | X | X | X | 0 | 1 | 1 | 38.400 kHz ± 38 Hz |
| X | X | X | X | X | 1 | 0 | 0 | 19.200 kHz ± 19 Hz |
| X | X | X | X | X | 1 | 0 | 1 | 9.6000 kHz ± 10 Hz |
| X | X | X | X | X | 1 | 1 | 0 | 4.8000 kHz ± 5 Hz |
| X | X | X | X | X | 1 | 1 | 1 | 2.4000 kHz ± 2 Hz |

(X denotes a don't care condition)



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3.3.8.5 UART register selection. UART registers shall be selected by latching the data in table XIII into U9 using the timing diagram of figure 2.

TABLE XIII. Data to select UART register.

| Register | Read or Load | P1-36 | P1-28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 |
|---|---|---|---|---|---|---|---|---|---|---|
| Transmitter holding | Load | 0 | X | X | X | 0 | 0 | 0 | 1 | 0 |
| Control | Load | 0 | X | X | X | 1 | 0 | 0 | 1 | 0 |
| Receiver holding | Read | 1 | X | X | X | 0 | 0 | 0 | 1 | 0 |
| Status | Read | 1 | X | X | X | 1 | 0 | 0 | 1 | 0 |

3.3.8.6 Control or transmitter holding register load. After selecting the control or transmitter holding register by the method of 3.3.8.5, these registers shall be loaded by applying the waveforms in figure 6.

3.3.8.7 RS-485 serial data out. The UART shall send out serial data by RS-485 communications protocol. Each RS-485 logic level 1 shall be a difference of no less than +0.2 Vdc but no greater than +12 Vdc between the U14 pins 5 and 9 and U14 pins 4 and 16. Each RS-485 logic level 0 shall be a difference of no greater than -0.2 Vdc but no less than -12 Vdc between U14 pins 5 and 9 and U14 pins 4 and 16. The voltage between U14 pins 5, 9 and P1-1, P1-2, P1-3, and P1-4 magnitude shall be less than $\pm$ 12 Vdc.

3.3.8.8 Data formats. The UART shall transmit and receive data with the formats shown in table XIV. After the UART control register has been selected, each format shall be initialized in the UART by applying the waveforms of figure 6.

TABLE XIV. Control register data for UART formats.

| Format | | P1- | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Data Bits | Stop Bits | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 |
| 7 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 |
| 8 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 0 |

3.3.8.9 Communications protocol select. After applying the waveforms of figure 2 with the fourth set of logic levels in table V, RS-232-C or RS-485 communications formats shall be selected by applying the waveforms in figure 5 to latch the data from the table XV into U12.

TABLE XV. Communications protocol select data.

| P1- | | | | | | | | Communications |
|---|---|---|---|---|---|---|---|---|
| 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | Protocol |
| X | X | X | X | X | X | 0 | 1 | None |
| X | X | X | X | X | X | 1 | 0 | RS-232-C |
| X | X | X | X | X | X | 1 | 1 | RS-485 |

(X denotes a don't care condition)



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3.3.8.10 RS-232-C serial data out. The UART shall send out the serial data by RS-232-C communications protocol. Each RS-232-C logic level 1 shall be in the range of -3 Vdc to -25 Vdc between P1-11, P1-12 and P1-15, P1-16. Each RS-232-C logic level 0 shall be in the range of +3 Vdc to +25 Vdc between P1-11, P1-12 and ground potential.

3.3.8.11 RS-485 serial data in. The UART shall receive serial data by RS-485 communications protocol. Each RS-485 logic level 1 shall be a difference of no less than +0.2 Vdc but no greater than +12 Vdc between P1-11, P1-12 and P1-15, P1-16. Each RS-485 logic level 0 shall be a difference of no greater than -0.2 Vdc but no less than -12 Vdc between P1-11, P1-12 and P1-15, P1-16. The voltage between U14 pins 5, 9 and P1-1, P1-2, P1-3 and P1-4 shall be no greater than +12 Vdc and no less than -12 Vdc.

3.3.8.12 Status and receiver holding registers read. The status and receiver holding registers shall be read by applying the waveforms of figure 7. The status register shall indicate when the transmitter shift register is empty, when the transmitter holding register is empty, and when data is available from the UART receiver holding register.

3.3.8.13 RS-232-C serial data in. The UART shall receive serial data by RS-232-C communications protocol. Each RS-232-C logic level 1 shall be in the range of -3 Vdc to -25 Vdc between the input line (P1-15 and P1-16) and ground potential. Each RS-232-C logic level 0 shall be in the range of +3 Vdc to +25 Vdc between the input line (P1-15 and P1-16) and ground potential.

3.3.8.14 Error recognition. The UART shall detect overrun, parity, and framing errors of incoming serial data and shall signal these errors by setting bits of the status register according to table XVI.

TABLE XVI. Status register flags.

| | P1- | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Flag | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 |
| Overrun error | X | X | 0 | 0 | 0 | 0 | 1 | 1 |
| Parity error | X | X | 0 | 0 | 0 | 1 | 0 | 1 |
| Framing error | X | X | 0 | 0 | 1 | 0 | 0 | 1 |

(X denotes a don't care condition)

3.3.8.15 RS-485 self test. The I/O PWA shall enable both the RS-485 driver and receiver at the same time to route serial data output from the UART back in as serial data input to verify communications sending and receiving capabilities.

3.3.9 Automated testing equipment (ATE) test. The PWA shall pass TACOM approved ATE test installed in TACOM approved ATE or equivalent.



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3.3.10 Soldering. Soldering shall meet or exceed the requirements of ANSI/IPC J-STD-001 (see 4.2).

3.4 Interface requirements. Each PWA shall accommodate the following requirements.

3.4.1 Overall envelope. Overall envelope of the PWA shall be in accordance with drawing 12259253.

3.5 Ownership and support requirements.

3.5.1 Interchangeability. Interchangeability tolerances should permit parts, subassemblies and assemblies to be used in their parent assemblies without regard to the source of supply or manufacturer. Parts, subassemblies and assemblies having the full range of dimensions and characteristics permitted by the specification governing the part, subassembly or assembly should be usable as replacement items without selection and without departure from the specified performance guidelines of the parent items.

3.5.2 Identification marking. Unless otherwise specified (see 6.2), identification marking shall be permanent and legible and shall as minimum include the following:

a. Nomenclature “PRINTED WIRING ASSEMBLY, I/O”
b. Military part number.
c. Manufacturer’s cage code and name.
d. Manufacturer’s serial numbers.
e. Contract number.
f. Symbol “US”.

3.6 Operating environment requirements. Each PWA shall operate under the following environmental conditions without damage or loss of performance.

3.6.1 Temperature.

3.6.1.1 High temperature (operating). The PWA shall be capable of operation without damage or malfunction at a high temperature of +125°F (52°C) (without solar radiation).

3.6.1.2 Low temperature (operating). The PWA shall be capable of operation without damage or malfunction at a low temperature of +20°F (-7°C). For ambient temperatures below +20°F (-7°C), the test set shall be operated in a shelter.

3.6.1.3 High temperature (nonoperating). The PWA shall be capable of being stored and transported in its case without damage at an ambient temperature of +160°F (71°C).



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3.6.1.4 Low temperature (nonoperating). The PWA shall be capable of being stored and transported in its case without damage at an ambient temperature of -60°F (-51°C).

3.6.2 Fungus. All nonmetallic materials that are not inherently fungistatic shall function in a fungus environment without adversely affecting the operation/performance of the PWA.

3.6.3 Shock. The PWA shall evidence no degradation of function and meet the requirements of 3.3 when subjected to the shock test.

3.6.4 Vibration. The PWA shall evidence no degradation of function and meet the requirements of 3.3 when subjected to the vibration test.

3.6.5 Burn-in. Unless otherwise specified (see 6.2), the PWA shall be subjected to and pass a "power on" burn-in prior to the inspections listed in 4.1.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspection (CI) (see 4.1.2).

4.1.1 First article inspection. First article inspection shall be performed on the initial production representative PWA of an order or when required by the contract. When a first article inspection is required (see 6.2 and 6.3), it includes all verifications listed in table XVII.

4.1.2 Conformance inspection. Conformance inspection shall include those examinations and tests from table XVII on all PWA produced as specified in contract.

TABLE XVII. Verification methods.

| Title | Requirement | Verification |
|---|---|---|
| Design, materials, & manufacturing processes | 3.2 | 4.2 |
| Operating requirements: | 3.3 | 4.2.3 |
| Input power | 3.3.1 | 4.2.3.1 |
| Logic levels | 3.3.2 | 4.2.3.2 |
| Input isolation | 3.3.3 | 4.2.3.3 |
| DCA/-15 Vdc source | 3.3.4 | 4.2.3.4 |
| Reset | 3.3.5 | 4.2.3.5 |
| BUS transceiver | 3.3.6 | 4.2.3.6 |



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TABLE XVII. Verification methods - Continued.

| Title | Requirement | Verification |
|---|---|---|
| Multiply/divide unit | 3.3.7 | 4.2.3.7 |
| Communications | 3.3.8 | 4.2.3.8 |
| ATE Test 1/ | 3.3.10 | 4.2.3.9 |
| Interface requirements | 3.4 | 4.2.4 |
| PWA overall | 3.4.1 | 4.2.4.1 |
| Ownership & support requirements | 3.5 | 4.2.5 |
| Interchangeability | 3.5.1 | 4.2.5.1 |
| Identification marking | 3.5.2 | 4.2.5.2 |
| Operating environment requirements | 3.6 | 4.2.6 |
| Temperature | 3.6.1 | 4.2.6.1 |
| Fungus | 3.6.2 | 4.2.6.2 |
| Shock | 3.6.3 | 4.2.6.3 |
| Vibration | 3.6.4 | 4.2.6.4 |

1/ Not required unless specified (see 6.2).

4.2 Verification methods. The types of verification methods included in this section are visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

4.2.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedures, to verify performance. See the contract for alternatives that replace verifications required by this specification.

4.2.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed under the following standard (room) ambient conditions:

a. Temperature: 77 ± 18°F (25 ± 8°C).
b. Relative humidity: Uncontrolled room ambient.
c. Atmospheric pressure: Site pressure.

4.2.3 Operating requirements verification.

4.2.3.1 Input power. To determine conformance to 3.3.1, the input current shall be measured with an ammeter.



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4.2.3.2 Logic levels.

4.2.3.2.1 CMOS Logic levels. To determine conformance to 3.3.2.1, the input and output CMOS logic levels shall be observed and verified with an oscilloscope.

4.2.3.2.2 TTL logic levels. To determine conformance to 3.3.2.2, the input and output TTL logic levels shall be observed and verified with an oscilloscope.

4.2.3.3 Input isolation.

4.2.3.3.1 Data BUS isolation. To determine conformance to 3.3.3.1:

a. Connect a 10K (where K = 1000) ± 10% ohm resistor to each of the bus board pins (P1-21 through P1-28). Terminate each of the 10 Kohm resistors with a 20K ± 10% ohm resistor to 5.00 V.
b. Apply 5.00 V to the connection between the 10 Kohm and the 20 Kohm resistors (the test input) connected to P1-21. Apply 0.00 V to the other seven test inputs. Verify that the voltage at P1-21 is not less than 4.5 V and the voltages at the other seven bus board pins are not more than 0.5 V.
c. Repeat part b for the seven remaining board pins P1-22 through P1-28.

4.2.3.3.2 J3-T, J3-U, IOF/7, MRD/, OEN-Q/, Q, CLEAR/, and TPB isolation. To determine conformance to 3.3.3.2.

a. Connect a 10K ± 10% ohm resistor to each of the board pins P1-11, P1-15, P1-35 through P1-38, P1-57, and P1-66.
b. Apply 5.00 V to the unterminated end (the test input) of the 10 Kohm resistor connected to P1-11. Apply 0.00 V to the other eight test inputs. (See 3.3.3.2 for proper P1-62 measurement.)
c. Verify that the voltage at P1-11 is not less than 4.5 V and the voltages at the other seven test inputs are not more than 0.5 V.
d. Repeat parts b and c for P1-15, P1-35 through P1-38, P1-57, and P1-66. (See 3.3.3.2 for proper P1-62 measurement.)

4.2.3.4 DCA/-15 Vdc source. To determine conformance to 3.3.4, apply voltages in the range of +0.40 Vdc to 32.00 Vdc to P1-42. Observe the voltages at P1-19 and P1-20.

4.2.3.5 Reset.

4.2.3.5.1 Power-up reset. To determine conformance to figure 1 and 3.3.5.1, apply +5 Vdc to P1-5 and P1-6 and observe the voltage at P1-68, U1 pin 21, U2 pin 2, U3 pin 2, U9 pin 1, U12 pin 13, and U20 pin 6.



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4.2.3.5.2 CLEAR/. To determine conformance to 3.3.5.2, apply a logic level 0 to P1-62 and observe the logic level at P1-68.

4.2.3.5.3 States during power-up reset or CLEAR/. To determine conformance to 3.3.5.3, maintain a logic level 0 at P1-68 and observe the logic levels at the pins specified in table II.

4.2.3.6 BUS transceiver.

4.2.3.6.1 Data in. To determine conformance to 3.3.6.1, verify that U4 pin 1 is at logic level 0. Apply the logic levels in table III to P1-21 through P1-28 and observe the corresponding logic levels in the table at U4 pins 2 through 9.

4.2.3.6.2 Extended I/O latch. To determine conformance to 3.3.6.2, verify that U4 pin 1 is at logic level 0. Apply the logic levels in table IV to P1-21 through P1-28 and latch them into U9 by applying the waveforms in figure 2. Verify that the corresponding logic levels in the table are present at U9 pins 2, 5, 7, 10, 12 and 15.

4.2.3.6.3 Extended I/O decoder. To determine conformance to 3.3.6.3, verify that U4 pin 1 is at logic level 0. Apply the logic levels in table V to P1-21 through P1-24 and latch them into U9 by applying the waveforms in figure 2. Verify that the corresponding logic levels in the table are present at U9 pins 10, 11, 12, 13, and 14.

4.2.3.6.4 BUS transceiver direction logic. To determine conformance to 3.3.6.4:

a. Apply a logic level 0 to P1-35 and a logic level 1 to P1-36. Verify that U4 pin 1 is at logic level 0.
b. Apply a logic level 1 to P1-35 and a logic level 1 to P1-36. Verify that U4 pin 1 is at logic level 0.
c. Apply logic level 0 to P1-35 and P1-36. Apply the first set of logic levels in table XVIII to P1-21 through P1-28 and latch them into U9 by applying the waveforms in figure 2. Apply a logic level 1 to P1-35 and P1-36. Verify that U4 pin 1 is at logic level 1.
d. Repeat part c for the next byte in table XVIII.

TABLE XVIII. Extended I/O logic levels.

| P1- | | | | | | | |
|---|---|---|---|---|---|---|---|
| 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 |
| X | X | X | X | 0 | 0 | 0 | 1 |
| X | X | X | X | 0 | 0 | 1 | 0 |



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4.2.3.7 Multiply/divide unit (MDU).

4.2.3.7.1 MDU crystal oscillator. To determine conformance to 3.3.7.1, observe the output of Y2 pin 8 with an oscilloscope.

4.2.3.7.2 MDU chip select logic. To determine conformance to 3.3.7.2:

a. Apply a logic level 0 to P1-35. Verify that U1 pin 23, U2 pin 11, and U3 pin 11 are at logic level 0.
b. Apply a logic level 1 to P1-35 and P1-57. Verify that U1 pin 23, U2 pin 11, and U3 pin 11 are at logic level 0.
c. Apply a logic level 1 to P1-35. Apply logic level 0 to P1-36 and P1-57. Verify that U1 pin 23, U2 pin 11, and U3 pin 11 are at logic level 1.
d. Apply a logic level 1 to P1-35 and P1-36. Apply a logic level 0 to P1-57. Verify that U1 pin 23, U2 pin 11, and U3 pin 11 are at logic level 1.

4.2.3.7.3 MDU register selection. To determine conformance to 3.3.7.3, apply the logic levels in table VIII at P1-21 through P1-26 and latch them into U9 by applying the waveforms in figure 2.

4.2.3.7.4 MDU register load. To determine conformance to 3.3.7.4:

a. Select the MDU control register using the method of 4.2.3.7.3.
b. Apply the logic levels in table XIX at P1-21 through P1-28 and load them into the MDU by applying the waveforms in figure 3.
c. Select the MDU X register using the method of 4.2.3.7.3 and repeat part b, once for the high byte and once for the low byte.
d. Select the MDU Y register using the method of 4.2.3.7.3 and repeat part b, once for the high byte and once for the low byte.
e. Select the MDU Z register using the method of 4.2.3.7.3 and repeat part b, once for the high byte and once for the low byte.

TABLE XIX. MDU register load data.

| MDU Register | High or Low Byte | Data at P1- 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 |
|---|---|---|---|---|---|---|---|---|---|
| Control | High | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 0 |
| X | High | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| | Low | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| Y | High | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| | Low | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| Z | High | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| | Low | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |



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4.2.3.7.5 MDU register read. To determine conformance to 3.3.7.5:

a. Apply the logic levels in table XX for the control register at P1-21 through P1-28 and load them into the MDU by applying the waveforms of figure 3.
b. Select the MDU status register using the method of 4.2.3.7.3.
c. Apply the waveforms of figure 4 to read the MDU status register. Verify that the logic levels match these given in table XX for the status register.
d. Select the MDU X register using the method of 4.2.3.7.3. Apply the waveforms of figure 4, once for the high order byte and once for the low order byte. Verify that the logic levels match those in table XIX for the X register.
e. Select the MDU Y register using the method of 4.2.3.7.3. Apply the waveforms of figure 4, once for the high order byte and once for the low order byte. Verify that the logic levels match those in table XIX for the Y register.
f. Select the MDU Z register using the method of 4.2.3.7.3. Apply the waveforms of figure 4, once for the high order byte and once for the low order byte. Verify that the logic levels match those in table XX for the Z register.

TABLE XX. MDU register read data.

| MDU Register | Read or Load | High or Low Byte | Data at P1- 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 |
|---|---|---|---|---|---|---|---|---|---|---|
| Control | Load | High | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| Status | Read | High | 0 | 0 | 0 | 0 | 0 | 0 | 0 | X |
| X | Read | High | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| | Read | Low | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| Y | Read | High | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| | Read | Low | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| Z | Read | High | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| | Read | Low | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |

X = Don't care.

4.2.3.7.6 Multiply with add. To determine conformance to 3.3.7.6:

a. Select the MDU control register using the method of 4.2.3.7.3.
b. Apply the logic levels in table XXI at P1-21 through P1-28 and load them into the MDU by applying the waveforms in figure 3.
c. Select the MDU X register using the method of 4.2.3.7.3 and repeat part b, once for the high byte and once for the low byte.
d. Select the MDU Y register using the method of 4.2.3.7.3 and repeat part b, once for the high byte and once for the low byte.
e. Select the MDU Z register using the method of 4.2.3.7.3 and repeat part b, once for the high byte and once for the low byte.
f. Select the MDU control register using the method of 4.2.3.7.3 and repeat part b.



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g. Apply the waveforms of figure 4 to read the MDU. Wait 17 µs. Verify that it matches the data in table XXI for the status register.
h. Select the MDU X register using the method of 4.2.3.7.3 and apply the waveforms of figure 4, once for the high byte and once for the low byte. Verify that the logic levels match those given in table XXI for the X register.
i. Select the MDU Y register using the method of 4.2.3.7.3 and apply the waveforms of figure 4, once for the high byte and once for the low byte. Verify that the logic levels match those given in table XXI for the Y register.
j. Select the MDU Z register using the method of 4.2.3.7.3 and apply the waveforms of figure 4, once for the high byte and once for the low byte. Verify that the logic levels match those given in table XXI for the Z register.

TABLE XXI. MDU register read data.

| Latch figure | Register | Read or Load | High or Low Byte | Data at P1-28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | Control | Load | High | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 0 |
| 4 | X | Load | High | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 |
| 4 | | Load | Low | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 4 | Y | Load | High | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| 4 | | Load | Low | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 4 | Z | Load | High | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 |
| 4 | | Load | Low | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 |
| 4 | Control | Load | High | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 |
| 5 | X | Read | High | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 |
| 5 | | Read | Low | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 5 | Y | Read | High | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 5 | | Read | Low | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 |
| 5 | Z | Read | High | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 5 | | Read | Low | 1 | 0 | 1 | 1 | 1 | 1 | 0 | 0 |
| 5 | Status | Read | High | 0 | 0 | 0 | 0 | 0 | 0 | 0 | X |

X = Don't care

4.2.3.7.7 Divide with overflow indication. To determine conformance to 3.3.7.7:

a. Repeat the process of 4.2.3.7.3 substituting the data in table XXII.
b. Select the MDU status register using the method of 4.2.3.7.3. Apply figure 4 to read the MDU and verify that it matches the data in table XXII for the status register.



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TABLE XXII. Data for MDU divide with overflow.

| Latch figure | Register | Read or Load | High or Low Byte | Data at P1- 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | Control | Load | High | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 0 |
| 4 | X | Load | High | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 4 | | Load | Low | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 |
| 4 | Y | Load | High | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 4 | | Load | Low | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 1 |
| 4 | Z | Load | High | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 4 | | Load | Low | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 1 |
| 4 | Control | Load | High | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 |
| 5 | X | Read | High | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 5 | | Read | Low | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 |
| 5 | Y | Read | High | X | X | X | X | X | X | X | X |
| 5 | | Read | Low | X | X | X | X | X | X | X | X |
| 5 | Z | Read | High | X | X | X | X | X | X | X | X |
| 5 | | Read | Low | X | X | X | X | X | X | X | X |
| 5 | Status | Read | High | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |

X = Don't care

4.2.3.8 Communications.

4.2.3.8.1 UART crystal oscillator. To determine conformance to 3.3.8.1, observe the output of Y1 pin 8 with an oscilloscope.

4.2.3.8.2 Crystal frequency division. To determine conformance to 3.3.8.2, observe the signals at U5 pins, 6, 5, 3, 2, 4, 13, 12 and 14 with an oscilloscope.

4.2.3.8.3 Frequency select latch. To determine conformance to 3.3.8.3, verify that U4 pin 1 is at logic level 0. Apply the logic levels in table X at P1-21 through P1-28 and latch them in by applying the waveforms in figure 2. Apply the logic levels in table XI to P1-21 through P1-28 and latch them into U7 by applying the timing diagram of figure 5. Verify that the corresponding logic levels in table XI are present at U7 pins 2, 7, and 10.

4.2.3.8.4 Frequency select. To determine conformance to 3.3.8.4, verify that U4 pin 1 is at logic level 0. Apply the logic levels in table XII to P1-21 through P1-23 and latch them into U7 by applying the timing diagram in figure 5. Verify that the corresponding frequencies in table XII are present in U6 pin 5.



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4.2.3.8.5 UART register selection. To determine conformance to 3.3.8.5, verify that U4 pin 1 is at logic level 0. Apply the logic levels in table XIII at P1-21 through P1-25 and latch them into U9 by applying the waveforms in figure 2.

4.2.3.8.6 Control and transmitter holding register load. To determine conformance to 3.3.8.6:

a. Select the control register using the method of 4.2.3.8.5.
b. Apply the logic levels in table XXIII at P1-21 through P1-28 and load them into the control register by applying the waveform in figure 6.
c. Select the transmitter holding register using the method of 4.2.3.8.5.
d. Apply the logic levels in table XXIII at P1-21 through P1-28 and load them into the transmitter holding register by applying the timing diagram in figure 6.

TABLE XXIII. UART load data.

| | P1- | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Register | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 |
| Transmitter holding | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| Control | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 |

4.2.3.8.7 RS-485 serial data out. To determine conformance to 3.3.8.7:

a. After performing the operations of 4.2.3.8.6, verify that the voltage of the signal at U14 pins 5 and 9 minus the voltage of the signal at U14 pins 4 and 16 conforms to the waveform in figure 8 while conforming to the limits given in 3.3.8.7.
b. Repeat the operations of 4.2.3.8.6 substituting the logic levels in table XXIV to load into the transmitter holding register. Verify that the voltage of the signal at U14 pins 5 and 9 minus the voltage of the signal at U14 pins 4 and 16 conforms to the waveform in figure 9 while conforming to the limits given in 3.3.8.7.

TABLE XXIV. UART serial data out.

| | P1- | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Register | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 |
| Transmitter holding | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |

4.2.3.8.8 Data formats. To determine conformance to 3.3.8.8:

a. Select the control register using the method of 4.2.3.8.5.
b. Latch the logic levels in table XIV for 8 data bits and 1 stop bit into U1 using the method of 4.2.3.8.6.
c. Load the transmitter holding register using the method and logic levels of 4.2.3.8.6 part d.



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d. Verify that the voltage of the signal at U14 pins 5 and 9 minus the voltage of the signal at U14 pins 4 and 16 conforms to the waveform in figure 10.
e. Repeat parts a through d using the logic levels in table XIV for 7 data bits and 1 stop bit. Verify the output with figure 8.

4.2.3.8.9 Communications protocol select. To determine conformance to 3.3.8.9:

a. Verify that U4 pin 1 is at logic level 0. Apply the fourth set of logic levels in table V and latch them into U9 by applying the timing diagram in figure 2. Apply the first set of logic levels to table XV at P1-21 and P1-22 and latch them into U12 by applying the timing diagram in figure 5.
b. Repeat the operations of 4.2.3.8.7 and verify that there are no signals present at P1-11, P1-12, P1-15 and P1-16.
c. Repeat part a for the RS-232-C in protocol table XV. Repeat the operations of 4.2.3.8.7 and verify that the signal between P1-11, P1-12 and P1-15, P1-16 conforms to the waveform in figure 11.
d. Repeat part a for the RS-485 protocol in table XV. Repeat the operations of 4.2.3.8.7 and verify that the signal between P1-11, P1-12 and P1-15, P1-16 conforms to the waveform in figure 8 and the limits given in 3.3.8.7.

4.2.3.8.10 RS-232-C serial data out. To determine conformance to 3.3.8.10:

a. Select RS-232-C protocol using the method of 4.2.3.8.9 part c.
b. Apply the first set of logic levels in table XIV at P1-21 through P1-28 and load them into the control register by applying the waveform in figure 6.
c. Select the transmitter holding register using the method of 4.2.3.8.5.
d. Apply the logic levels in table XXIV at P1-21 through P1-28 and load them into the transmitter holding register by applying the timing diagram in figure 6.
e. Verify that the voltage of the signal at U13 pin 4 conforms to the waveform in figure 11 and the limits given in 3.3.8.10.

4.2.3.8.11 RS-485 serial data in. To determine conformance to 3.3.8.11:

a. Select RS-485 communications protocol using the method of 4.2.3.8.9 part d.
b. Load the control register for the 7 data bits and 1 stop bit format using the method of 4.2.3.8.8.
c. Apply the waveform of figure 8 as a difference between voltages at P1-11, P1-12 (positive) and P1-15, P1-16 (negative).
d. Verify that the waveform at U1 pin 20 conforms to the CMOS logic level requirements of 3.3.2.1 and conforms to the waveform of figure 8.



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4.2.3.8.12 Status and receiver holding registers read. To determine conformance to 3.3.8.12:

a. Select the status register using the method of 4.2.3.8.5. Repeat 4.2.3.8.11 and during the application of the waveform apply figure 7 to read the status register. Verify that the logic levels observed at P1-21 through P1-28 are those in table XXV for part a.
b. Apply the timing diagram of figure 7 to read the status register after the waveform in figure 8 has been applied. Verify that the logic levels observed at P1-21 through P1-28 are those in table XXV for part b.
c. Select the receiver holding register using the method of 4.2.3.8.4. Read the receiver holding register by applying the timing diagram in figure 7 and verify that the logic levels at P1-21 through P1-28 are 01010101.
d. Apply the timing diagram of figure 7 to read the status register and verify that the logic levels at P1-21 through P1-28 are those listed in table XXV for part d.
e. Select RS-232-C communications protocol using the method of 4.2.3.8.9 part a.
f. Apply the waveform in figure 11 and P1-11, P1-12 and P1-15, P1-16.
g. Read the transmitter holding register using the method of part c and verify that the logic levels are those in table XXIV. Read the status register using the method of part d and verify that the logic levels are those in table XXV for part d.

TABLE XXV. UART Status and receiver holding register read.

| Register | P1-28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 |
|---|---|---|---|---|---|---|---|---|
| Status Part a | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| Status Part b | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| Receiver holding | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| Status Part d | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |

4.2.3.8.13 RS-232-C serial data in. To determine conformance to 3.3.8.13:

a. Select RS-232-C communications protocol using the method of 4.2.3.8.9 part c.
b. Load the control register for the 7 data bits and 1 stop bit format using the method of 4.2.3.8.8.
c. Apply the waveform of figure 11 as a voltage level at P1-11, P1-12 with P1-15, P1-16 at ground potential.
d. Verify that the waveform at U1 pin 20 conforms to TTL logic level requirements of 3.3.2.1 and conforms to the waveform of figure 11.

4.2.3.8.14 Error recognition. To determine conformance to 3.3.8.14:

a. Select RS-485 communications protocol using the method of 4.2.3.8.9.



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b. Load the control register for the 7 data bits and 1 stop bit format using the method of 4.2.3.8.8.
c. Apply the waveform of figure 8 as a difference between voltages at P1-11, P1-12 (positive) and P1-15, P1-16 (negative). Immediately apply the waveform of figure 8 again.
d. Read the status register using the method of 4.2.3.8.5 and verify that the logic levels are those listed in table XXVI.
e. Repeat part c by applying the waveform in figure 12. Repeat part d and verify the second set of logic levels in table XXVI.
f. Repeat part c by applying the waveform in figure 13. Repeat part d and verify the third set of logic levels in table XXVI.

TABLE XXVI. UART error recognition data.

| | P1- | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Register | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 |
| Status Part d | X | X | 0 | 0 | 0 | 0 | 1 | 1 |
| Status Part e | X | X | 0 | 0 | 0 | 1 | 0 | 1 |
| Status Part f | X | X | 0 | 0 | 1 | 0 | 0 | 1 |

(X denotes a don't care condition)

4.2.3.8.15 RS-485 self test. To determine conformance to 3.3.8.15:

a. Select RS-485 communications protocol using the method of 4.2.3.8.9.
b. Load the control register for the 7 data bits and 1 stop bit format using the method of 4.2.3.8.8.
c. Apply the logic levels in table XXVII at P1-21 through P1-28 and latch them into U9 by applying the timing diagram in figure 2. Apply a logic level 1 to P1-21 and latch it into U8 pin 2 by applying the timing diagram in figure 5.
d. Load the transmitter holding register using the method and logic levels of 4.2.3.8.6 part d.
e. Read the receiver holding register using the method of 4.2.3.8.5 and verify that the logic levels are those in table XXVII for the transmitter holding register.

TABLE XXVII. RS-485 self test set up data.

| P1- | | | | | | | |
|---|---|---|---|---|---|---|---|
| 26 | 27 | 26 | 25 | 24 | 23 | 22 | 21 |
| X | X | X | X | 0 | 1 | 0 | 1 |

(X denotes a don't care condition)

4.2.3.9 ATE Test. To determine conformance to 3.3.9, install the PWA into a TACOM approved fixture (such as BTF-2500-93), on TACOM approved, ATE equipment (such as HP 3062 with TACOM approved software and instructions, such as BSW-25000930).



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4.2.4 Interface requirements verification.

4.2.4.1 PWA overall envelope inspection. Insert the PWA into the present existing location of the test set, and verify its envelope and interfacing conformance.

4.2.5 Ownership and support requirements verification.

4.2.5.1 Interchangeability inspection. To determine conformance to 3.5.1, the interchangeability between different part, subassemblies, and assemblies having the same military part number shall be verified.

4.2.5.2 Identification marking. Verify the presence of the required PWA identification marking.

4.2.6 Operating environment requirements verification. The electrical performance tests specified in 4.2.3 through 4.2.3.8.5 shall be conducted prior to and at the completion of each environmental test. Verification of performance during high temperature and low temperature environmental tests shall be demonstrated by the tests specified in 4.2.3.3, 4.2.3.4, 4.2.3.5, 4.2.3.6, 4.2.3.7, and 4.2.3.8.

4.2.6.1 Temperature. To determine conformance to 3.6.1, the PWA shall be tested as follows:

4.2.6.1.1 High temperature (operating). To determine conformance to 3.6.1.1, the PWA shall be subjected to high temperature test in accordance with method 501.3, procedure II (hot category), of MIL-STD-810, or equivalent (see 4.2.1), and tested to 3.3.

4.2.6.1.2 Low temperature (operating). To determine conformance to 3.6.1.2, the PWA shall be subjected to low temperature test in accordance with method 502.3, procedure II, of MIL-STD-810, or equivalent (see 4.2.1), and tested to 3.3.

4.2.6.1.3 High temperature (nonoperating). To determine conformance to 3.6.1.3, the PWA shall be subjected to high temperature test in accordance with method 501.3, procedure I, of MIL-STD-810, or equivalent (see 4.2.1).

4.2.6.1.4 Low temperature (nonoperating). To determine conformance to 3.6.1.4, the PWA shall be subjected to low temperature test in accordance with method 502.3, procedure I, of MIL-STD-810, or equivalent (see 4.2.1).

4.2.6.2 Fungus. To determine conformance to 3.6.2, the PWA shall be subjected to the applicable fungus test in accordance with method 508.4 of MIL-STD-810 (28 day test), or equivalent (see 4.2.1).



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4.2.6.3 Shock. To determine conformance to 3.6.3, the PWA shall be subjected to the shock test specified by method 516.4, procedure I, 40g peak of MIL-STD-810D, or equivalent (see 4.2.1), and tested to 3.3 after shock test.

4.2.6.4 Vibration. To determine conformance to 3.6.4, the PWA shall be subjected to a vibration test using a composite of figure 14 through 16. These figures represent the cargo environment at the cargo bed of a composite of tactical wheeled vehicles, the 5-ton M813 truck and the 12-ton M127 semitrailer. The data include differing vehicle loading conditions traversing over specially designed courses ranging from paved highway to off road conditions at various vehicle speeds. The spectrum is broad band random with peaks and notches at various discrete frequency bands. Break points are provided for establishing the spectrum shape. The test duration is 500 miles. The PWA shall be tested to 3.3 after the vibration test.

4.2.6.5 Burn-in. When required (see 6.2), the burn-in shall consist of three temperature cycles and a six gravity unit (g) root mean square (rms) random vibration as follows:

a. Temperature cycle. Each temperature cycle shall consist of 30 minutes at -60°F (-51°C) and 30 minutes at plus (+) 125°F (52°C). The rate of temperature change between temperature extremes shall be not less than 5°F (2.8°C) per minute. Power shall be applied to the PWA for not less than 30 minutes from 20°F (7°C) through the completion of each high-temperature soak.
b. Random vibration. Following all temperature cycling, the PWA shall be hard mounted to the vibration table so that the axis of vibration is perpendicular to the plane of the PWA. The vibration shall be at room ambient temperature, for a period of 10 minutes, as shown in figure 17.

NOTE: PWAs being procured for simultaneous installation in VTMs being procured do not have to be subjected to “burn-in” provided the VTMs are subjected to the 16-hour “power on” burn-in.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department or Defense Agency, or within the Military Department’s System Command. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.



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## 6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The PWA contains a Universal Asynchronous Receiver/Transmitter (UART; U1) with supporting logic and analog circuitry to facilitate external communications. Transmission and reception baud rate is software selectable, with a range from 150 to 19.2K bits per second. Communication can be in either RS-232-C or RS-485 format through P1-11 (board pins should be designated P1-XX, where XX is pin number), P1-12, and P1-15, P1-16. Software selected circuitry will route RS-485 output back as input for verification. The I/O PWA contains two cascaded Multiply/Divide Units (MDU's, U2 and U3) that are capable of an unsigned 16 bit by 16 bit multiply operation giving a 32 bit result. A 32 bit by 16 bit divide operation, giving a 16 bit result and a 16 bit remainder, is also available. Data is transferred on a 8 bit bus (P1-21 through P1-28). A bus transceiver (U4) isolates the data bus on the PWA from the data bus in the STE/ICE-R system. A -15 volt DCA source that cannot be pulled above +0.4 volts is available at P1-42. The PWA is military unique because it is designed specifically to be a component of the Simplified Test Equipment/Internal Combustion Engine - Reprogrammable (STE/ICE-R) Vehicle Test Meter (VTM), and to withstand environmental conditions which exceed commercial requirements.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. PWA drawing number, revision letter, date, and PWA part number (see 3.2 and 3.5.1).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. If first article inspection is required (see 3.1).
e. If logic levels should be other than as specified (see 3.3.2).
f. If marking is other than as specified (see 3.5.2).
g. If burn-in is not required (see 3.6.5).
h. Whether ATE test is required (see table XVII footnote).
i. Packaging requirements (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously



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acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, Government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table XVII and their designated frequency based on a risk assessment for the procurement.

6.5 Subject term (key word) listing.

Automated Testing Equipment (ATE) test
Bus Transceiver
Crystal oscillator
Multiply/Divide Unit (MDU)
RS-232-C
RS-485
Simplified Test Equipment/Internal Combustion Engine-Reprogrammable (STE/ICE-R)
Vehicle Test Meter (VTM)
Universal Asynchronous Receiver/ Transmitter (UART)

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

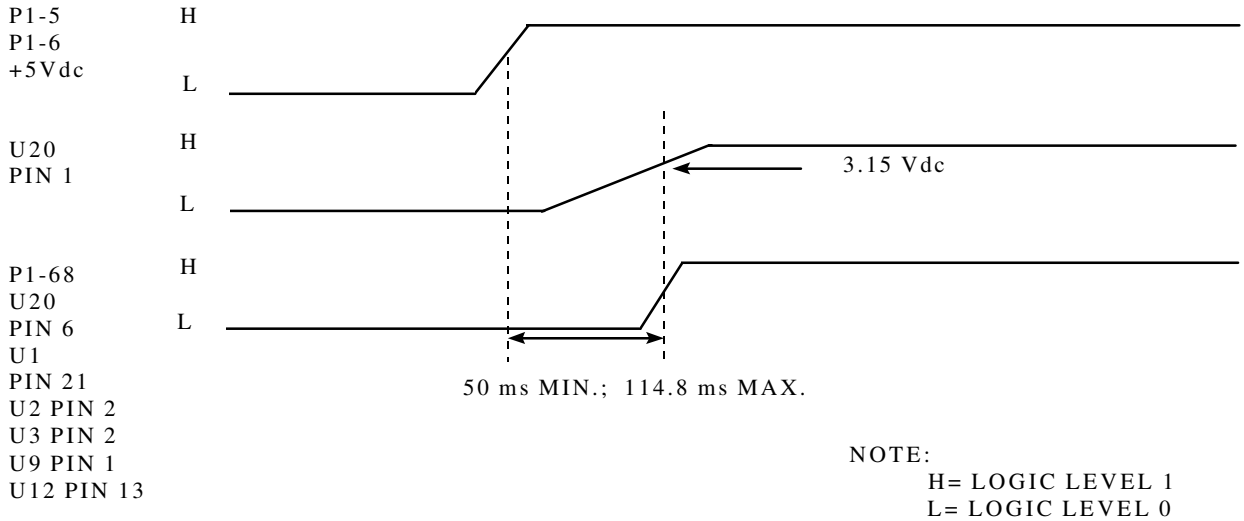



FIGURE 1. Power-up reset and CLEAR/ timing diagram.



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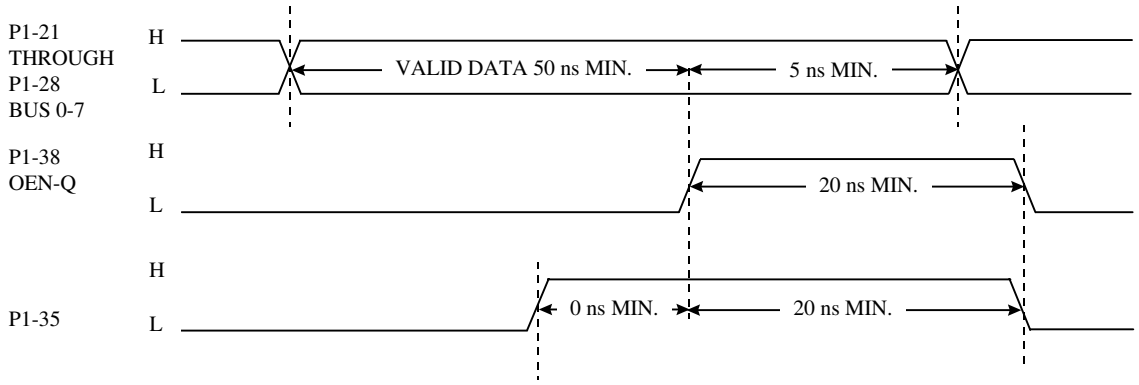
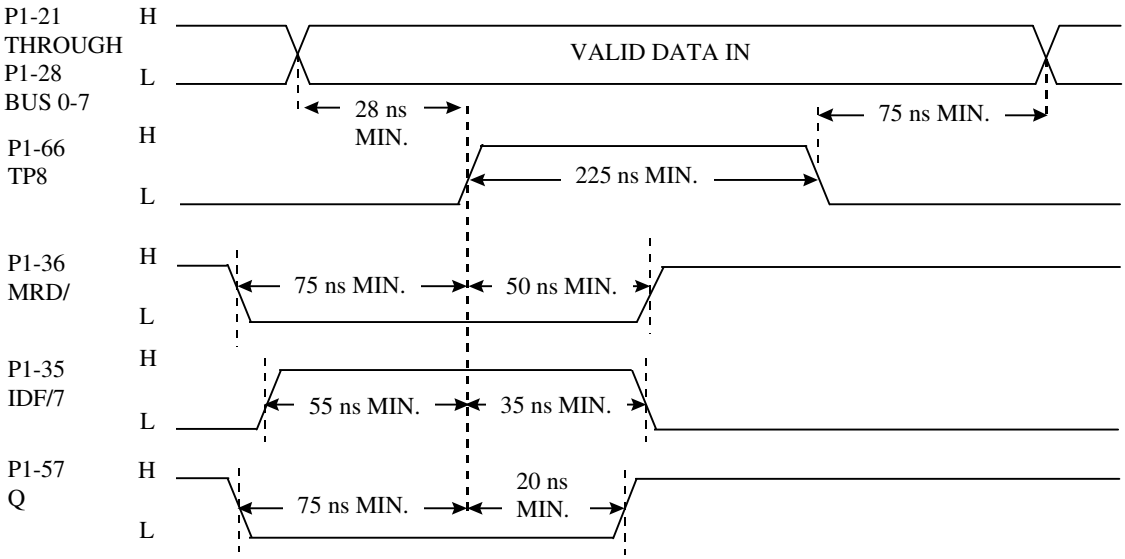



FIGURE 2. Timing diagram to latch data into U9.




NOTE:
H= LOGIC LEVEL 1
L= LOGIC LEVEL 0

FIGURE 3. Timing diagram to load MDU registers.

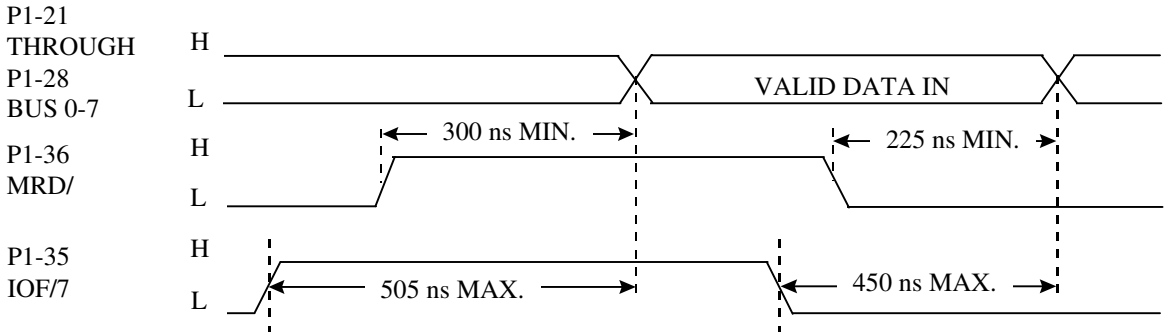
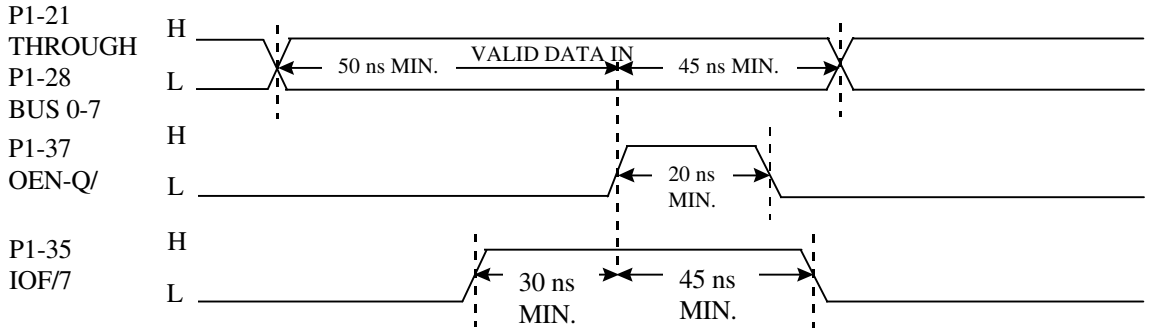



FIGURE 4. Timing diagram for reading MDU registers.



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NOTE:
H= LOGIC LEVEL 1
L= LOGIC LEVEL 0

FIGURE 5. Timing diagram to latch data into U7, U8 or U12.

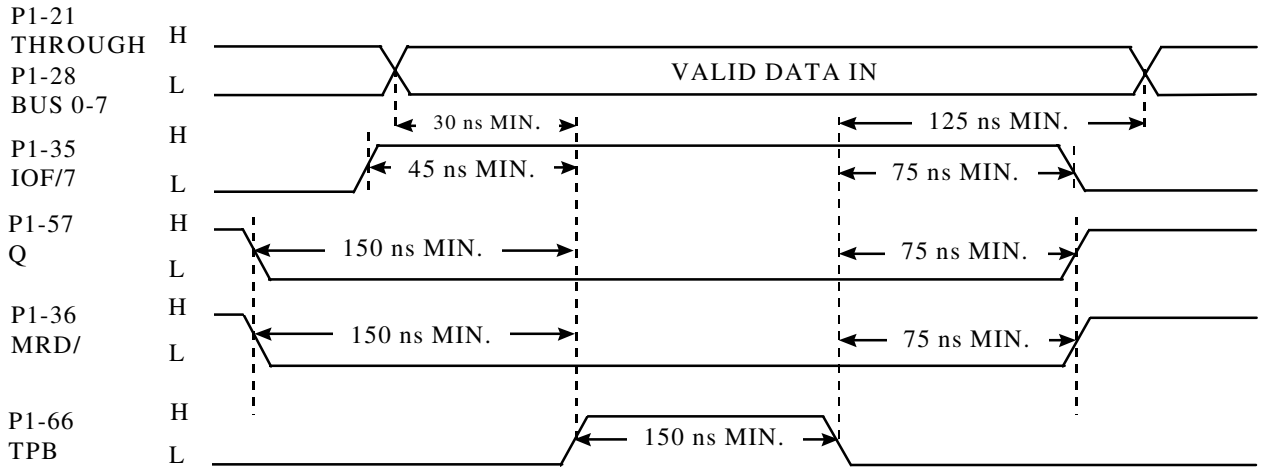



FIGURE 6. Timing diagram to read UART control and transmitter holding registers.

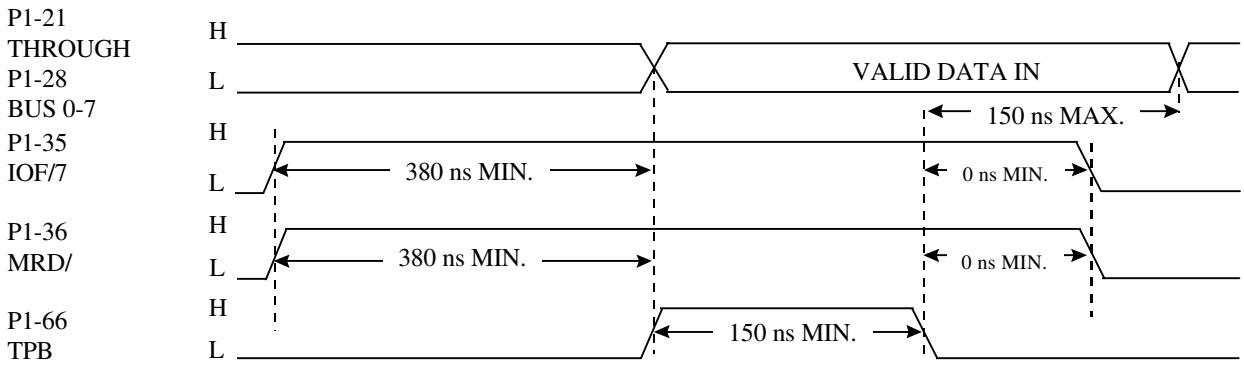



FIGURE 7. Timing diagram to read UART status and receiver holding registers.



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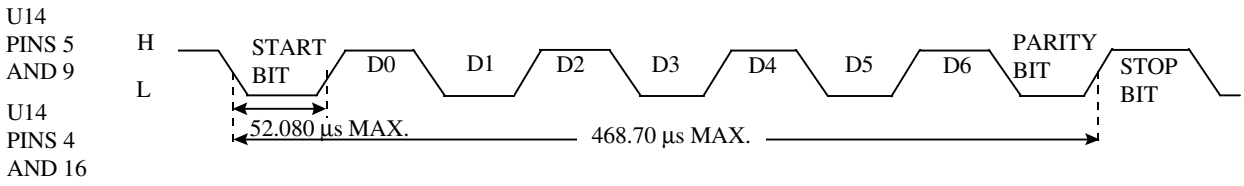



FIGURE 8. Serial data for RS-485 communications "55" with 7 data bits, 1 even parity bit, and 1 stop bit for input and output.

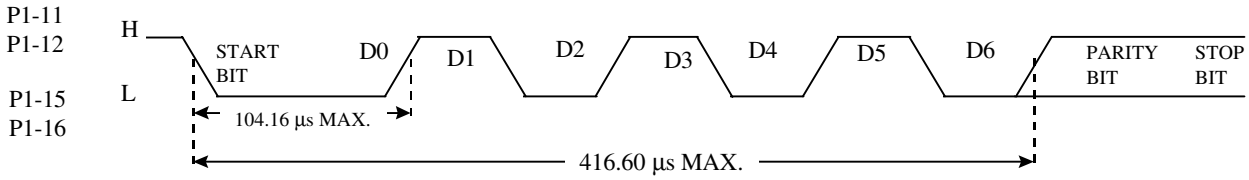
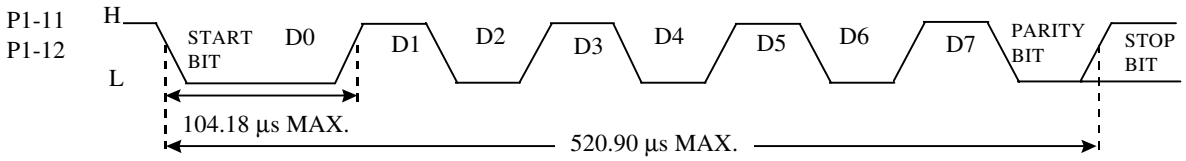



FIGURE 9. Serial data for RS-485 communications "24" with 7 data bits, 1 even parity bit, and 1 stop bit for input and output.




NOTE:
H= LOGIC LEVEL 1
L= LOGIC LEVEL 0

FIGURE 10. Serial data for RS-485 communications "AA" with 8 data bits, 1 even parity bit, and 1 stop bit for input and output.

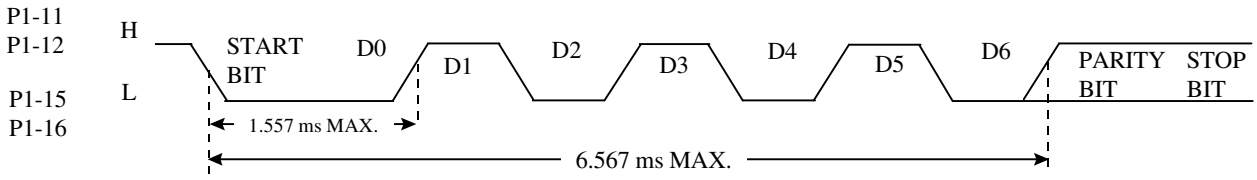



FIGURE 11. Serial data for RS-232 communications "2A" with 7 data bits, 1 even parity bit, and 1 stop bit for input and output.



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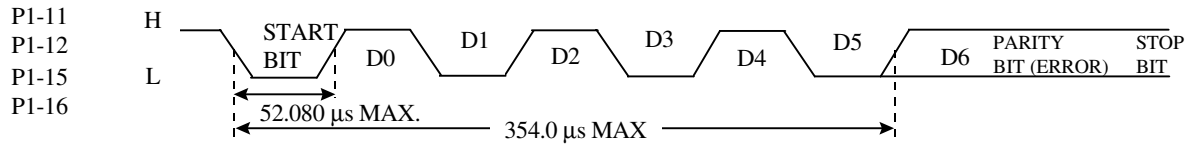



FIGURE 12. Serial data for RS-485 “55” with 7 data bits, 1 stop bit, and parity error for input.

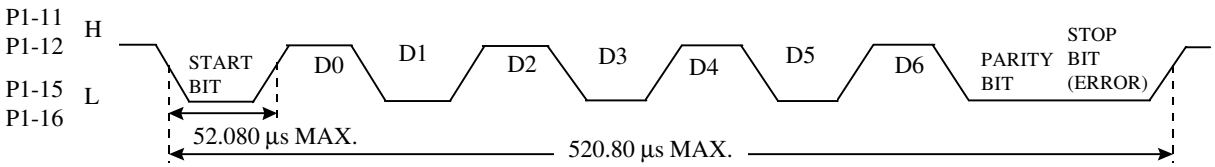
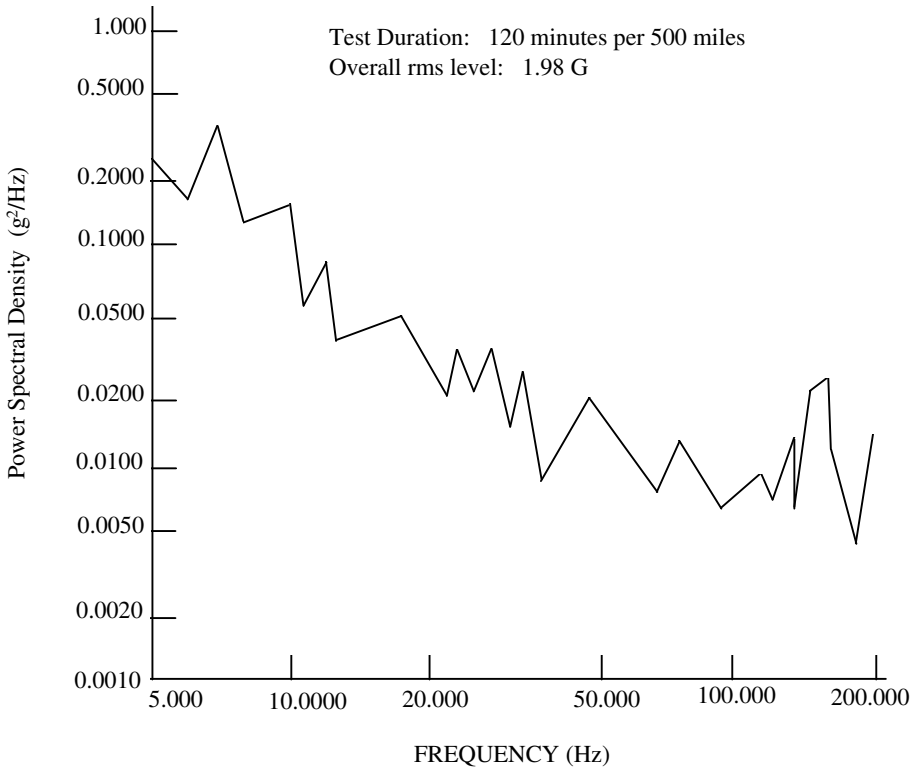



FIGURE 13. Serial data for RS-485 “55” with 7 data bits, 1 even parity bit, and framing error for input.

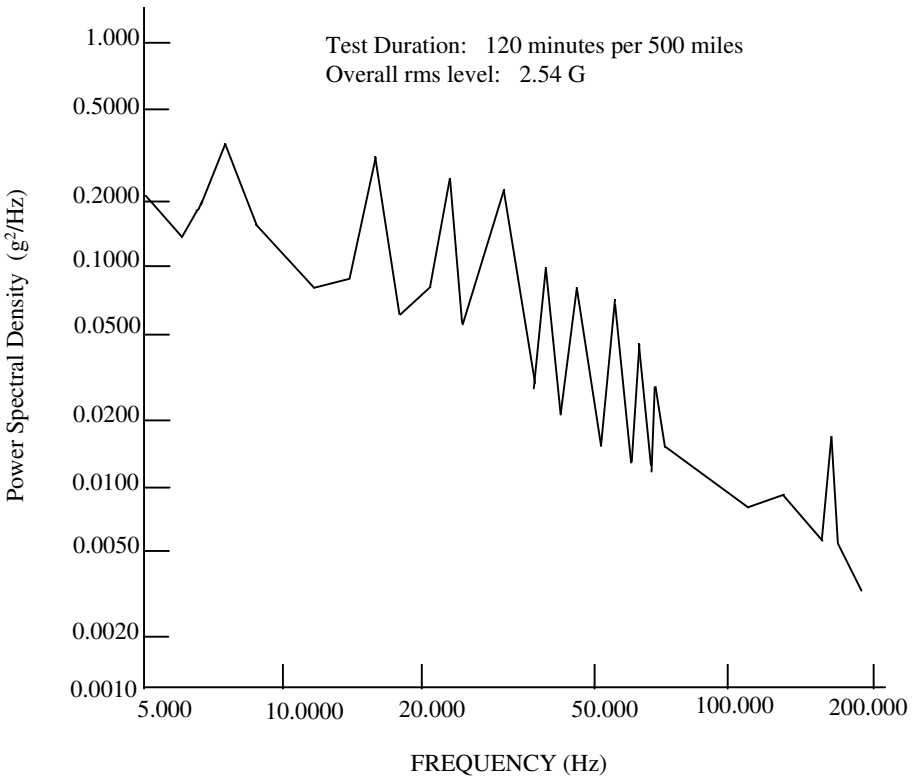



| BREAKPOINTS | |
|---|---|
| FREQ | PSD VALUE |
| 5 | 0.23840 |
| 6 | 0.16840 |
| 7 | 0.38230 |
| 8 | 0.14870 |
| 10 | 0.16840 |
| 11 | 0.05930 |
| 12 | 0.09750 |
| 13 | 0.04090 |
| 18 | 0.05370 |
| 23 | 0.01940 |
| 24 | 0.03520 |
| 26 | 0.02310 |
| 28 | 0.03520 |
| 32 | 0.01670 |
| 33 | 0.03110 |
| 37 | 0.00810 |
| 49 | 0.02190 |
| 63 | 0.00700 |
| 73 | 0.01150 |
| 98 | 0.00660 |
| 120 | 0.00970 |
| 123 | 0.00680 |
| 138 | 0.01340 |
| 140 | 0.00760 |
| 146 | 0.02490 |
| 159 | 0.02820 |
| 163 | 0.01300 |
| 188 | 0.00480 |
| 200 | 0.01340 |

FIGURE 14. Basic transportation, composite tactical wheeled environment, vertical axis.



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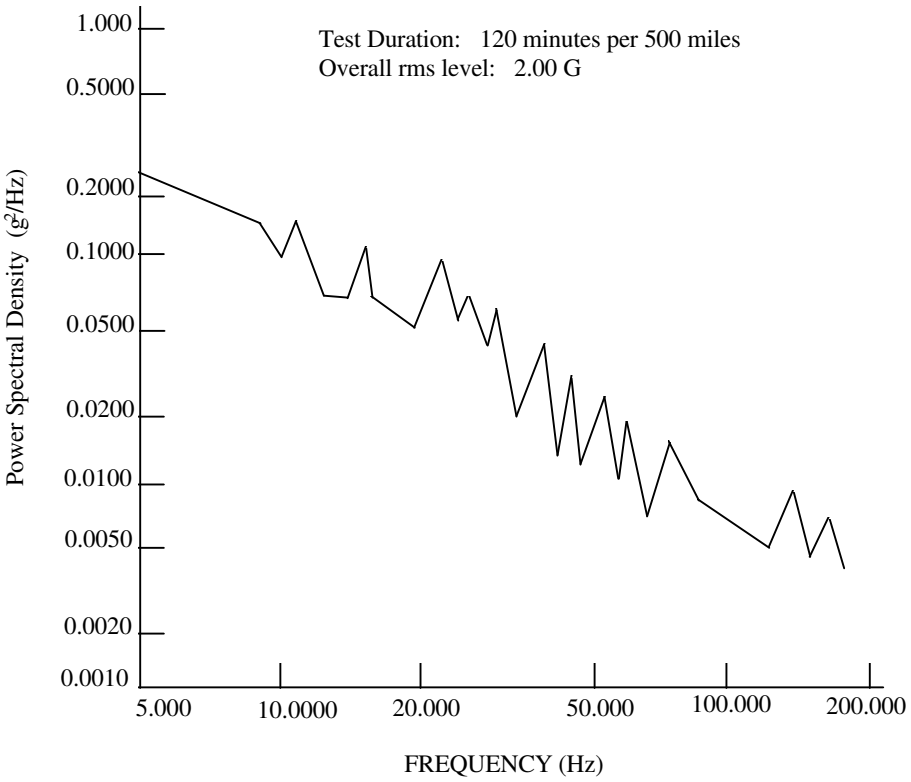



| BREAKPOINTS | |
|---|---|
| FREQ | PSD VALUE |
| 5 | 0.22130 |
| 6 | 0.13140 |
| 7 | 0.20540 |
| 8 | 0.38230 |
| 9 | 0.15630 |
| 12 | 0.07990 |
| 15 | 0.08830 |
| 16 | 0.32940 |
| 19 | 0.06080 |
| 22 | 0.09280 |
| 24 | 0.26340 |
| 27 | 0.05640 |
| 32 | 0.22690 |
| 36 | 0.02820 |
| 40 | 0.11600 |
| 44 | 0.02090 |
| 48 | 0.08190 |
| 53 | 0.01480 |
| 56 | 0.07420 |
| 60 | 0.01270 |
| 64 | 0.04980 |
| 65 | 0.01100 |
| 68 | 0.03260 |
| 69 | 0.01480 |
| 114 | 0.00740 |
| 132 | 0.00830 |
| 160 | 0.00510 |
| 169 | 0.01710 |
| 173 | 0.00550 |
| 200 | 0.00310 |

FIGURE 15. Basic transportation, composite tactical wheeled environment, longitudinal axis.



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| BREAKPOINTS | |
|---|---|
| FREQ | PSD VALUE |
| 5 | 0.25490 |
| 10 | 0.13030 |
| 11 | 0.09670 |
| 12 | 0.13360 |
| 13 | 0.06500 |
| 15 | 0.06340 |
| 16 | 0.11230 |
| 17 | 0.06500 |
| 22 | 0.04590 |
| 24 | 0.08980 |
| 26 | 0.05070 |
| 28 | 0.06830 |
| 30 | 0.03670 |
| 32 | 0.05880 |
| 38 | 0.01920 |
| 40 | 0.03860 |
| 44 | 0.01290 |
| 48 | 0.02940 |
| 50 | 0.01140 |
| 56 | 0.02460 |
| 60 | 0.00960 |
| 63 | 0.01880 |
| 70 | 0.00660 |
| 78 | 0.01500 |
| 93 | 0.00730 |
| 137 | 0.00460 |
| 145 | 0.00910 |
| 162 | 0.00430 |
| 176 | 0.00660 |
| 200 | 0.00360 |

FIGURE 16. Basic transportation, composite tactical wheeled environment, transverse axis.



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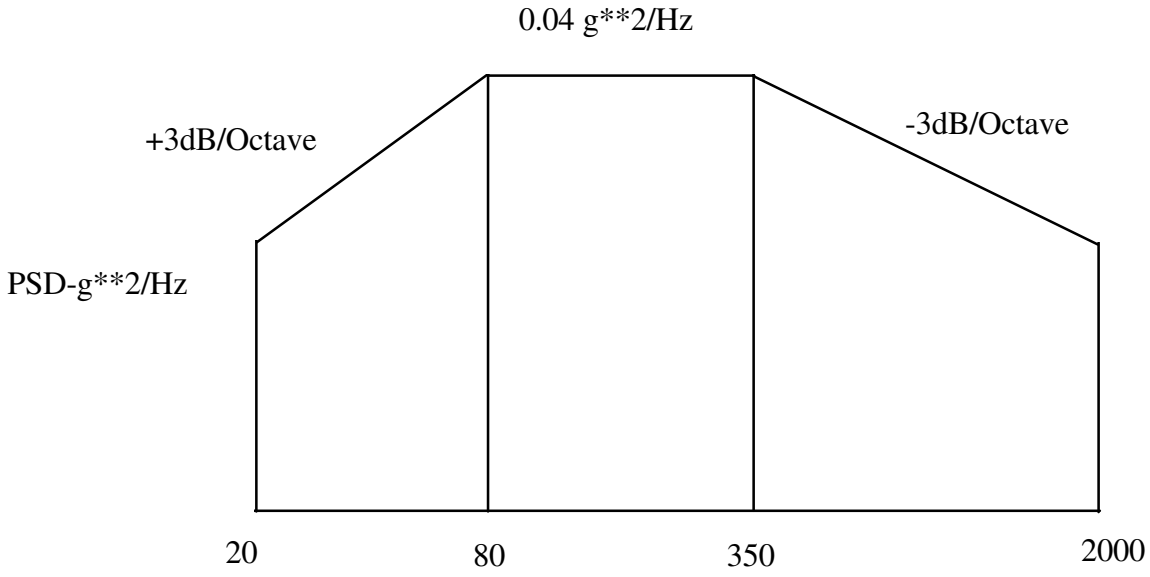



FIGURE 17. Random vibration spectrum.

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Army - AT

Preparing Activity:
Army - AT

(Project 5998-0073)



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**3. DOCUMENT TITLE**
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|---|---|
| | (810) 574-8745 786-8745 |
| c. ADDRESS *(Include Zip Code)*<br>Commander<br>U.S. Army Tank-automotive and Armaments Command<br>ATTN: AMSTA-TR-E/BLUE<br>Warren, MI 48397-5000 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403<br>Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |



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